

Memory Lifecycle: The LLM-Wiki Closed Loop

Bidirectional feedback loop crystallizing volatile runtime traces into compounding evergreen knowledge

STAGE 1

Session Loadout

Awaken & Context Injection

- <300 Token budget pack
- Project scope & active ADRs
- Recent L2 decision logs
- Fuzzy-matched concepts

STAGE 2

Runtime Offload

AST Trace Truncation

- Pipe to `k0maru offload`
- 99.44% Token reduction
- Extract error signatures
- Generate Mermaid state chart

STAGE 3

Trace-to-Skill

Pattern Distillation

- Heuristic error extractors
- Go/Node/Python/Rust models
- Auto-generate troubleshooting
- Direct export to skills/

STAGE 4

Session Flush

Atomic Crystallization

- Adaptive convention sniffer
- Anti-collision file writer
- WikiLinks graph synthesis
- Instant SQLite cache sync

STAGE 5

Compounding

Evergreen Long-Term Memory

- Obsidian / LLM-Wiki storage
- 1-hop bidirectional boost
- Hybrid BM25 + ONNX vectors
- Zero daemon, zero lock-in

Continuous Knowledge Enrichment: Crystallized skills & logs actively feed into future session loadouts